

OUTCOME-BASED

PART- C

19. Course Description

Second Year Second Semester		
Course Code: CSE 3101	Year: Third	Semester: First
Course Title: Operating Systems		
Course Status: Core		
Credit: 3.0		
Prerequisite(s): None		
Rationale	The course is essential for understanding the core principles that govern modern computing systems. It introduces students to critical concepts such as process management, memory allocation, file systems, and security, providing a strong foundation for system design and resource optimization. The primary objectives of the course are to equip students with the ability to understand how operating systems function, analyze system performance, and apply OS concepts to solve real-world problems. By combining theoretical knowledge with practical skills, the course prepares students to analyze, develop, and manage efficient operating systems, making them capable of addressing real-world challenges in software development and system administration. This course is fundamental for careers in computing and forms the basis for advanced studies in computer science and engineering.	

Week	Course Contents	CLOs
	Section A	
1	Introduction to Operating Systems: Evolution and goals of OS, Definition and types of operating systems, OS structure, System calls, and OS services.	1
2	Process Management I: Process concept, process states, process control block (PCB), context switching, and system calls for process management.	2
3, 4	Process Management II: Threads and multithreading models, CPU scheduling, and scheduling criteria and algorithms	3
5	Case Study and Review for Class Test-1. Class Test-1	3
	Section B	
6	Process Synchronization I: Synchronization mechanisms such as semaphores, mutexes, and monitors.	3, 4

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7	Process Synchronization II: Classical synchronization problems (Producer-Consumer, Dining Philosophers), deadlocks (conditions, prevention, avoidance, detection, and recovery).	3, 4
8	Memory Management I: Memory hierarchy and management, swapping, contiguous memory allocation, paging, and segmentation.	1
9	Memory Management II: Virtual memory concepts, demand paging, page replacement algorithms (FIFO, LRU, Optimal), and thrashing.	2
10	Case Study and Review for Class Test-2. Class Test-2	3
Section C		
11	File Systems: File concepts, access methods, directory and disk structure, and file system mounting and sharing, file allocation methods (contiguous, linked, indexed), free space management, file system implementation, and performance.	1, 3
12	Storage Management: Disk structure, disk scheduling algorithms (FCFS, SSTF, SCAN, C-SCAN), RAID levels, and disk management, including formatting, interrupts, direct memory access (DMA)	4
13	Security and Protection: Security threats, vulnerabilities, attack vectors, protection mechanisms such as access control, encryption, authentication, and includes case studies on OS security (Linux, Windows).	1, 4
14	Case Study and Review for Final Examination. Class Test-3	3

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Course Learning Outcomes (CLOs)	Upon completion of this course the students will be able to:		Mapping with PLOs
	CLO1	Identify the structure and functioning of various types of operating systems.	PLO1
	CLO2	Describe the fundamental concepts and principles of operating systems.	PLO1
	CLO3	Analyze the performance and efficiency of operating system components, such as scheduling algorithms, resource allocation, and security mechanisms.	PLO2
	CLO4	Apply system calls, threading, and synchronization techniques to develop solutions for real-world operating system problems.	PLO7

Mapping CLOs with the Teaching-Learning and Assessment Strategy

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture, Team Teaching	Quiz and Class Test
CLO2	Lecture, Presentation, Group Discussion	CT and Final Exam
CLO3	Lecture, Problem Solving, Group Discussion	CT, Assignment, Final Exam
CLO4	Lecture, Problem Solving	CT, Assignment, Final Exam

Learning Materials

Recommended Readings	Operating System Concepts (10th Edition) by Abraham Silberschatz, Peter B. Galvin, Greg Gagne Modern Operating Systems (4th Edition) by Andrew Tanenbaum, Herbert Bos
Supplementary Readings	Operating Systems (3rd Edition) by Harvey M. Deitel, Paul J. Deitel, David R. Choffnes Operating Systems: Internals and Design Principles (8th Edition) by William Stallings